

KRISHNA REDDY SAI THARUN KANCHAM

kkrsai97@gmail.com | 9407583899 | Dallas, Tx

PROFESSIONAL SUMMARY

- **8 years** of progressive experience starting as an Integration Engineer in the telecom industry, transitioning to Data Analyst, and currently serving as an **AWS Data Engineer**, delivering scalable solutions across Telecom, Pharma and Logistics domains.
 - Experienced with software development life cycle, architecting scalable platforms, object-oriented programming, database design and Agile methodologies.
 - Skilled in building robust Big Data pipelines on AWS EMR and Databricks, applying DevOps best practices with CI/CD automation and implementing rigorous unit and regression testing frameworks for ETL validation.
 - Experience in writing data processing frameworks for large scale applications.
 - Hands on experience in using Spark ecosystem components like S3, RDS, Snowflake, REST, Kafka.
 - Involved in the CI/ CD pipeline management for managing the weekly releases.
 - Experience in consuming data from Kafka into spark micro batches.
 - Worked on spark application tuning and resource allocations based on use case.
 - Familiar with JSON based REST Web services and Amazon Web services.
 - Automated the continuous integration and deployments using Jenkins and AWS Cloud Templates deployment services (Lambda).
 - Experience building **CI/CD** Pipeline to automate the code release process using Integration tools like Git, GitHub, Jenkins.
 - Experience in setting up **CI/CD** pipeline integrating various tools with Jenkins to build and run Terraform jobs to create infrastructure in AWS.
 - Experience in project deployment using Jenkins and using web services like Amazon Web Services (AWS) EC2, Cloud Formation Templates, AWS S3 and Cloud watch.
 - Applied **Big Data processing techniques** using tools like Apache Spark, Hive, and HBase for large-scale ETL, real-time ingestion, and analytics in cloud-based data pipelines
-

TECHNICAL SKILLS

- **Languages** : Python, PySpark, SQL.
 - **Tools**: GitHub, Terraform, Docker, JIRA, DBeaver, SQL Server Management Studio (SSMS), Databricks, Snowflake
 - **Databases**: MySQL, SQL Server, Oracle DB, Dynamo DB.
 - **ETL/Orchestration**: AWS Glue, AWS Step Functions, AWS Data Pipelines, AWS Sagemaker
 - **AWS Stack**: Lambda, DynamoDB, SNS, SQS, API Gateway, AWS CloudFormation, AWS Event Bridge, Amazon Redshift, RDS, AWS Kinesis Data Stream, AWS Firehose, S3, AWS Cloud Watch, AWS Media Live, Athena, AWS Textract , DMS, Amazon Connect, Amazon Lex, AWS Well Architected Review, AWS MSK, AWS Data Pipeline, AWS Quick Sight, AWS Kendra, AWS Bedrock .
 - **Big Data Frameworks**: AWS EMR, Apache Spark, Databricks.
 - **Streaming & Messaging**: Kafka, AWS Kinesis, AWS Firehose, Structured Streaming
 - **CI/CD** : Jenkins
 - **Visualization**: Tableau, QlikView.
-

EDUCATION

Master of Science in Business Analytics

University of North Texas – Denton, TX 2021 – 2023

Bachelor of Technology in Electronics and Communication Engineering

Amrita University – Kerala, India 2014 – 2018

CERTIFICATIONS

AWS Certified Data Engineer - Associate (2024)

AWS Certified Solution Architect - Associate (2023)

AWS Certified Cloud Practitioner – Fundamental (2023)

EXPERIENCE

AWS Data Engineer, BizCloud Experts, Dallas, USA

June 2023 – present

- As a Data Engineer, I worked on **Data Pipeline Modernization, 214 Data Integration, Omni Live Logistics** for Omni Logistics and **B2B Data Integration** for RNDC.
- Designed and maintained scalable **ETL pipelines** to process orders, shipment, tracking, and warehouse data using **AWS Glue, S3, and Redshift**.
- Handled infrastructure provisioning and job monitoring for Glue crawlers, Spark jobs, and triggers, ensuring high availability and minimal failure rates in production environments.
- Engineered and maintained scalable Big Data pipelines using **PySpark** on **AWS Glue, AWS EMR, and Databricks**, processing millions of records for logistics and supply chain data.
- Led Databricks-based data engineering projects to build end-to-end pipelines leveraging **Autoloader, Delta Lake, Notebooks, Workflows, and Jobs**, orchestrated on dedicated and autoscaling **Spark clusters**.
- Implemented Delta Lake bronze → silver → gold architecture with support for schema evolution, versioning, and real-time analytics using time travel and audit-friendly data snapshots.
- Managed end-to-end data migration activities using **AWS DMS**, enabling seamless replication of on-premise and RDS data into Amazon Redshift, Snowflake, and S3 for analytics use cases.
- Developed and orchestrated **data workflows on AWS EMR**, leveraging Spark clusters for large-scale data transformations, batch processing, and data lake optimizations.
- Designed and implemented streaming pipelines with **watermarking, late event tolerance, and backpressure-aware transformations** using **Structured Streaming in Databricks**.
- Managed AWS Infrastructure components such as **EMR clusters, S3 Data Lakes, Lambda functions, and CloudWatch** for scalable, cost-effective big data processing.
- Ensured secure and governed data movement through **IAM policies**, encryption standards, and **CloudWatch** monitoring for ETL job performance and error tracking.
- Proficient in building **data pipelines using AWS Glue, AWS Data Pipeline, and AWS DMS**, enabling efficient ETL workflows.
- Ingested streaming data from **Kafka topics into Spark** for real-time processing and transformation in Databricks and EMR environments.
- Integrated **PyTest-based unit and regression tests** to validate ETL logic, row-level transformations, and schema conformance across development and production stages.
- Interacting with business owners to gather both functional and technical requirements.
- Spearheaded **Data Pipeline Modernization**, replacing legacy architectures with more efficient pipelines to support growing data needs Omni Legacy Account.
- Supported the modernization of legacy ETL workflows by transitioning them to **serverless architectures** using Glue, Lambda, and Step Functions.
- Worked in Agile methodology and participated in all Agile ceremonies. Worked on understanding the project requirements and created the Integration solution design. Worked on Databases, created tables, views, temporary tables, stored procedures and reporting tables for OMNI Logistics.
- Documenting the business requirements and framing the business logic for the ETL process.
- Use **Agile** methodology for SDLC and utilize scrum meetings for creative and productive work.
- Design, develop, and test **Informatica** mappings, workflows, worklets, reusable objects, SQL queries, and Shell scripts to implement complex business rules.
- Demonstrated expertise in crafting and optimizing complex **SQL queries** for data extraction, transformation, and reporting for Omni Logistics.
- Built data pipelines and databases tailored to AMS's business requirements, ensuring data accuracy and scalability.
- Led the development and relaunch of the Fields Manager website with a complete backend overhaul, including database redesign, optimized database management, and implementation of core business logic to enhance functionality and performance.
- Implemented Jenkins CI/CD pipelines for Databricks and Glue job deployments with integration into AWS EMR-based workflows.
- **Developed serverless applications** using **AWS Lambda, API Gateway, DynamoDB, S3, and Event Bridge**, reducing operational overhead and improving system scalability.
- Developed and deployed **serverless applications using CI/CD pipelines**, ensuring automated testing, security validation, and continuous deployment.
- Developed an **end-to-end B2B integration platform using AWS** to replace legacy ERP and TIBCO systems for RNDC as a pilot modernization initiative.

- Developed a **machine learning model** for Fields Manager that auto-detects breaks in live sports streams, auto-generates replays, and inserts commercials using Amazon Rekognition.

Environment: Linux, Windows, SQL, Serverless, AWS (Glue, Data Pipeline, DMS, Lambda, API Gateway, DynamoDB, S3, Event Bridge, Redshift, EMR), Databricks, PySpark, Agile, Scrum, GitHub, CI/CD, JSON, ETL, AWS B2B, TIBCO, API Development, Machine Learning, Data Warehousing, Node JS.

Data Engineer, Cigna Healthcare, USA

Oct 2021 – May 2023

- Analyzed **pharmacy claims data**, focusing on the **parent-child relationship mapping** claims data and developing transformation logic for migration efforts.
- Designed layouts, data requirements, and data pipelines for **Supplemental Discount Claims**, ensuring seamless data integration and processing.
- Collaborated with multiple teams, including **Facets, Treasury, HCPM, Gateway, and Everest**, to gather data requirements and organize data for smooth project execution.
- Demonstrated strong knowledge of claims data components like **Source, DAL, Master, Extract, LZ, UCL, HCPM, and Packing Slip**, ensuring accurate data handling.
- Created **visualization reports** using tools like Tableau to support data analysis, data mapping, and data cleansing efforts, enabling insights for key stakeholders.
- Created a mapping document that outlines the sources mapped to the targets.
- Created, optimized, reviewed, and executed Complex SQL queries to validate transformation rules used in source to target mappings/source views, and to verify data in target tables.
- Created views to select data from the existing SQL Server databases.
- Extensively worked on database performance tuning techniques and modifying the complex join statements.
- Created **Functional Spec & Technical Spec documentation** & also documented the issues found in the end-to-end testing.
- Extensively worked with DBA's during the performance testing phase for our database
- Identifying **Bottlenecks, Optimizing SQL, Reducing Unnecessary Caches** etc.
- Designed and developed scripts for administrative tasks like backup's, tuning and periodically refreshing the test databases from the production databases.
- Supporting the application in Production environment by monitoring the **ETL process** every day during the nightly loads.
- Developed triggers and stored procedures for data verification and processing.
- Involved in extensive data validation by writing several complex SQL queries and involved in **back-end testing** and worked with data quality issues.
- Led **test case preparation** for monitoring data validation during the **Oracle DB upgrade** from version 12c to 19c, ensuring data integrity throughout the process.
- Executed data migration tasks, including **migrating tables, schemas, and databases**, while developing and applying transformation logic for smooth transitions.
- Utilized **Tableau** to create data visualizations for bank employees, delivering insightful reports to aid decision-making.
- Processed **batch files** into data warehouses and data marts, supporting structured storage and retrieval for business intelligence.

Environment: Linux, Windows, SQL Server, Tableau, ETL, Data Warehousing, Data Marts, Database Performance Tuning, Data Mapping, Toad for Oracle, Jira, Mapping Transformations, REST, JSON, Pyspark

Data Analyst, Ericsson, Bangalore, INDIA

Oct 2016 – Aug 2021

- Developed and optimized **ETL pipelines** for processing large-scale **telecom network and customer data** for T-Mobile.
- **Designed and optimized Oracle-based Data Warehouses**, implementing **partitioning, indexing, and query tuning** to improve performance.
- Created **custom dashboards and reports** using **Power BI, Tableau** for network analytics.
- **Implemented data modeling** techniques for **customer churn prediction, network congestion forecasting, and fraud detection**.
- Utilized **Ericsson Cloud Data Storage and Processing Solutions** to maintain high availability of telecom data.
- Expertise in Batch Processing and Batch Provisioning.
- Worked as an Integration Engineer in Ericsson dealing Provisioning systems for multiple network providers across the globe namely TMO, Sprint, SimInn, GCI, LMT, Thuraya, TDC, IEC.
- Mainly deals with Installation, Integration, Testing (Manual and Automation) of Ericsson product EDA (Ericsson Dynamic Activation) over various regions with Installations were made Native, ECEE, Open stack, Containers.
- Collaborated with a team of 4, Developed an Automated Test Suite with 343 Test Cases and reduced the efforts to 37min

from 4-5 working days.

- Expertise in System Testing, Node Testing, Functional Testing, UAT,E2E Testing, Acceptance, Sanity, Regression, SIT.
- Hands on experience with Integration activities from EDA Node to vUDC, SAPC, MTAS, ENUM, ENM, HLR, MNP, HSS, CUDB.
- Driven Multiple Maintenance windows as EDA Engineer for EDA, Cassandra Upgrade Activities.
- CICD activities on EDA tool, Handful of experience in development of LLD, MOP and other documents.
- Successfully Delivered Upgrade activities in Production Environments and Lab Environments (72) within 54 days for T-Mobile in United States.

Environment: Linux, Windows, Oracle, SQL, Tableau, EECp , ETL, Data Warehousing, Data Modeling, Network Analytics, OpenStack, Containers, CI/CD, Cassandra, Robo Framework, Integration Engineering, Production & Lab Environment Deployment, Upgrade Activities, Maintenance Windows, Low-Level Design (LLD), Method of Procedure (MOP), Telecom Data Processing.